Restaurant Recommendation System

A Project Report
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Abstract

In the modern digital age, finding a suitable restaurant has become more complex because of the large number of choices available to consumers. Traditional food recommendation platforms mainly rely on generic 5-star rating systems, which often fail to capture the unique preferences and tastes of individual users. The project titled **"Restaurant Recommendation System"** is designed to overcome this limitation by developing an intelligent machine learning-based model that suggests restaurants using semantic analysis, customer feedback, and cuisine-related features.

For this study, a detailed dataset consisting of **51,717 restaurant records from Bangalore**, sourced through **Zomato**, is collected and systematically pre-processed. **Exploratory Data Analysis (EDA)** is performed to extract valuable insights such as highly rated restaurants, frequently preferred cuisines, and rating patterns across different locations in the city. A **Content-Based Filtering** technique is implemented using the **TF-IDF (Term Frequency–Inverse Document Frequency)** vectorizer to analyze textual reviews, while **Cosine Similarity** is used as a mathematical measure to identify similarity between restaurant profiles.

To enhance usability and practical implementation, the final similarity model is optimized and deployed through an interactive web application built with the **Flask** framework. The application allows users to enter the name of a preferred restaurant and instantly receive a personalized set of similar restaurant recommendations. This improves the dining selection process while showcasing the practical application and effectiveness of data science techniques in recommendation systems.

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1. Introduction

The hospitality and food sector has grown significantly in recent years. With platforms such as Zomato offering a wide range of dining choices, users often struggle with “choice overload.” A common 4.5-star rating system does not effectively differentiate between varied dining experiences, such as a vibrant fast-food restaurant and a peaceful fine-dining setting. Modern consumers increasingly seek personalized recommendations that suit their preferences in taste, atmosphere, and overall dining experience.

The **Restaurant Recommendation System** is designed to address this challenge by replacing conventional trial-and-error decision-making with a data-driven solution. Through the analysis of large amounts of restaurant reviews and related information, the system applies artificial intelligence techniques to understand the deeper context, features, and characteristics associated with each restaurant.

This system provides practical advantages in three major real-world situations:

1. For Restaurant Visitors:

It allows users to identify dining options that match their individual preferences, improving user satisfaction while reducing the time spent searching for suitable restaurants.

2. For Restaurant Owners:

It supports businesses in reaching the most relevant customer base, improving audience targeting, increasing customer visits, and enhancing retention rates.

3. For Food Delivery Platforms:

By integrating such a recommendation system, platforms can offer more precise suggestions to users, leading to greater engagement and a higher frequency of orders.

2. Project Overview

This project is a complete intelligent web-based application developed to operate as a culinary recommendation engine. Its architecture follows a well-structured data science pipeline that transforms raw restaurant data into meaningful recommendations through an interactive and user-friendly interface.

The process starts with **Data Collection**, where a large dataset containing restaurant information—such as restaurant names, pricing, cuisines, locations, and customer reviews—is obtained. This data then goes through **Data Cleaning and Pre-processing**, which involves handling missing values, eliminating duplicate records, and standardizing textual content. After this, **Feature Engineering** is performed to merge important attributes, creating a more detailed and informative representation of each restaurant profile.

The processed data is then fed into the **Recommendation Engine**, where **Natural Language Processing (NLP)** methods such as TF-IDF are utilized to transform textual reviews into numerical vectors. These vectors are analyzed using **Cosine Similarity**, which allows the system to calculate the relationship between restaurants based on their features and customer opinions. To enhance performance, the similarity matrix is optimized and stored, enabling quick retrieval when handling real-time recommendation requests.

At the user end, interaction takes place through a responsive **Web Application** interface. Users enter or select a restaurant of their choice, and the backend processes the request by matching it with the precomputed similarity matrix. Based on this analysis, the system produces a ranked list of similar restaurants, ensuring personalized and accurate recommendations.

Furthermore, the system is developed with scalability and usability as key considerations. It supports fast processing speeds, a modular structure for future enhancements (such as collaborative filtering or hybrid recommendation models), and the possibility of integrating user feedback mechanisms to continuously improve recommendation quality and system accuracy.

3. Objectives

- **Build Knowledge in Content-Based Filtering:** Develop a strong understanding of creating recommendation systems that depend on item-related features instead of user interaction history.
- **Carry Out Advanced Data Preprocessing:** Gain the ability to clean, organize, and transform real-world datasets, with emphasis on processing unstructured text using methods such as tokenization, stop-word elimination, and text normalization.
- **Perform Exploratory Data Analysis (EDA):** Utilize data visualization techniques to identify patterns, trends, and relationships within the dataset, supporting improved decision-making and effective feature selection.
- **Apply Suitable Machine Learning Methods:** Implement relevant algorithms such as TF-IDF for extracting features and Cosine Similarity for evaluating relationships among restaurant profiles.
- **Develop and Deploy a Full-Stack Web Application:** Transform the analytical model into a complete web-based system using Flask, while integrating frontend technologies like HTML, CSS, and JavaScript to deliver an interactive user interface.
- **Improve System Performance and Usability:** Focus on optimizing the recommendation engine for quicker response times while maintaining scalability and efficiency for practical, real-world applications.

4. Project Initialization and Planning Phase

a) Define Problem Statement
Consumers often face difficulty while choosing dining places because of the overwhelming number of generalized reviews available online. Existing platforms mostly depend on overall ratings, which may not accurately reflect an individual user's specific food preferences. There is a strong need for an automated system that can analyze deeper characteristics of a restaurant—such as its menu offerings, ambiance, and customer feedback—to recommend dining experiences that are genuinely similar.

b) Project Proposal (Proposed Solution)
The proposed solution is an NLP-based recommendation system. By processing restaurant reviews through the TF-IDF algorithm, the system captures the unique textual characteristics or "vocabulary" associated with each restaurant. Through a web-based interface, users can search for a restaurant they already prefer, while the backend engine identifies and returns the top matching restaurants based on textual similarity as well as category-based factors such as pricing and overall ratings.

c) Initial Project Planning
Before development began, the required environment and technology stack were established:

• **Development Environment:** Anaconda and Anaconda Navigator were used to write, test, and manage the code efficiently and securely.

• **Data Handling and Computation:** pandas and NumPy were selected for efficient matrix operations and data frame processing.

• **Visualization Tools:** Matplotlib, Seaborn, and Plotly were chosen to create informative charts, graphs, and distribution visualizations.

• **Machine Learning:** scikit-learn was used for its reliable feature extraction methods and similarity measurement algorithms.

• **Web Deployment:** Flask was selected as the lightweight framework for server-side routing and rendering HTML pages.

5. Data Collection and Preprocessing Phase

a. Data Collection Plan and Raw Data Sources Identified
The model was built using the widely recognized **"Zomato Bangalore Restaurants"** dataset sourced from Kaggle. This dataset contains **51,717 records** with **17 attributes**. The major features chosen for developing the model include:

• **name:** Represents the name of the restaurant.

• **rate:** Indicates the restaurant's overall rating out of 5.

• **cuisines:** Specifies the food categories offered, listed in comma-separated format.

• **Approx cost (for two people):** Represents the estimated dining cost for two people.

• **Reviews list:** Contains a large collection of textual reviews submitted by customers.

b. Data Quality Report
A detailed assessment of the dataset was performed using functions such as pandas .info() and .isnull().sum(). The analysis identified several common real-world data challenges, including missing (NaN) values in multiple columns. In addition, important numerical fields such as ratings and costs were stored as strings (for example, "4.1/5" and "1,200"), which made them unsuitable for direct computation. The review text also contained excessive punctuation and special symbols, making cleaning necessary before analysis.

c. Data Preprocessing
To prepare the dataset for analysis and model development, several preprocessing techniques were applied:

• **Removing Unnecessary Features:** Columns that did not contribute to recommendations, such as URL, phone, and dish liked, were removed to improve efficiency.

• **Handling Missing Values and Duplicates:** Rows containing null values were removed, and duplicate entries were eliminated to preserve consistency in the dataset.

• **Feature Formatting:** The approx_cost(for two people) column was renamed as **cost**. The "/5" was removed from the rate column, commas were removed from the cost column, and both fields were converted into float numerical data types for computation.

• **Mean Rating Scaling:** A custom script was created to calculate the mean rating for each unique restaurant. These ratings were then standardized using MinMaxScaler with a feature range of 1 to 5.

• **Text Cleaning and Standardization:** Text in the reviews list and cuisines columns was converted to lowercase, punctuation was removed, and common stop words were eliminated using NLP methods to improve the quality of analysis.

d. Exploratory Data Analysis (EDA) and Visualization

Data visualization was important in identifying patterns and extracting useful insights:

• **Most Popular Restaurants:** A bar chart showed that Cafe Coffee Day, Onesta, and Empire Restaurant have a strong presence across Bangalore.

• **Rating Trends:** A distribution analysis indicated that a large number of restaurants have ratings above 3.5, reflecting a generally positive dining quality.

• **Top Restaurants and Cuisines:** Additional visualizations highlighted highly rated restaurants such as Byg Brewski Brewing Company and showed that popular cuisine combinations include North Indian cuisine, Indian Chinese cuisine, and Fast food based on frequent word-pair analysis.

6. Model Development Phase

a. Model Selection Report

For this text-focused recommendation system, a **Content-Based Filtering** method was chosen. Unlike collaborative filtering, which depends on user activity and behavior patterns, content-based filtering makes recommendations using the characteristics and attributes of the items themselves.

To process rich textual information, the **TF-IDF (Term Frequency–Inverse Document Frequency)** technique was applied. TF-IDF serves as a statistical method that measures the importance of a word within a document relative to a collection of documents. It reduces the weight of commonly occurring words while assigning greater significance to unique and descriptive terms that better represent a restaurant's atmosphere, cuisine, and identity.

b. Initial Model Training, Validation and Evaluation Report

The cleaned `reviews_list` data was passed through the **TfidfVectorizer** to transform textual information into numerical vectors. After vectorization, **Cosine Similarity** was used to measure the similarity between restaurants. Cosine similarity is a metric used to determine how closely two documents are related, regardless of their length.

The similarity between Restaurant A and Restaurant B was computed using the following formula:

$$similarity(A, B) = \frac{A.B}{\|A\| \times \|B\|} = \frac{\sum_{i=1}^{n} A_i \times B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \times \sqrt{\sum_{i=1}^{n} B_i^2}}$$

Using the linear kernel applied to the TF-IDF matrix, the system generated a large similarity matrix containing the similarity score of every restaurant compared with all other restaurants in the dataset. This enabled the recommendation engine to identify and rank the most relevant restaurants for any selected input.

7. Model Optimization and Tuning Phase

a. Tuning Documentation

To improve the recommendation engine for final deployment, specific optimization logic was incorporated into the recommendation function:

• **N-Gram Range Optimization:** The TF-IDF vectorizer was tuned using ngram_range=(1,2), allowing the model to capture both individual words and meaningful two-word phrases (for example, "fine dining"). This improved the system's ability to understand more contextual relationships within reviews.

• **Result Filtering:** When a search query is submitted, the algorithm retrieves the top 30 most similar restaurant indexes. It then removes duplicate branches with identical restaurant names to avoid recommending multiple outlets of the same franchise. After filtering, the remaining results are ranked according to Mean Rating, and the top 10 highest-rated and most similar restaurants are returned to the user.

b. Final Model Selection Justification

This pipeline—combining TF-IDF with Cosine Similarity—was selected as the final model because it is computationally efficient and well-suited for sparse textual data. In addition, incorporating ranking logic based on Mean Rating and Cost improves the quality of recommendations, ensuring users receive not only similar restaurant options but also the highest-quality alternatives available.

8. Results

The mathematical model was successfully transitioned from a static Jupyter Notebook environment into a dynamic, full-stack web application using Python's Flask framework. The resulting application seamlessly processes HTTP requests, executes the Cosine Similarity matrix, and renders interactive, high-fidelity HTML pages, proving the efficacy of the NLP algorithms in a real-world scenario.

8.1 System Initialization and Execution

To achieve the final results and operate the recommendation engine, the system follows a strict initialization, training, and deployment protocol:

- **Environment Preparation:** The deployment environment is configured using Python 3.x. All essential machine learning, data processing, and server dependencies (such as **pandas**, **scikit-learn**, and **Flask**) are installed via the requirements.txt.
- **Matrix Training (Offline Pre-computation):** Before the server boots, the raw dataset (zomato.csv.zip) is processed via the Restaurant_Recommendation_System.ipynb notebook. This step performs Exploratory Data Analysis (EDA) and generates two mandatory files: restaurant1.csv (cleaned dataset) and restaurant.pkl (pre-computed TF-IDF mathematical matrix).
- **Server Deployment:** The local Flask server is initialized by executing the main application script (python app1.py).
- **Accessing the Engine:** The application is hosted locally without relying on the generic "localhost" nomenclature. The user accesses the fully deployed engine by navigating their preferred web browser directly to the address: http://127.0.0.1:5000/.

8.2 Interface Experience and Output

The system is designed as an interactive premium web platform named **Vellore & Vine**, featuring a dark luxury-inspired interface with a refined visual identity. The design language combines a deep black background theme with gold accents (#C5A059) and elegant typography using **Cormorant Garamond** and **Montserrat**, creating a sophisticated and immersive user experience.

• Landing Interface (index.html)
The landing page serves as the introductory interface and establishes the brand identity through the concept of **"Neural Palate"** and algorithm-driven dining intelligence. It includes a premium hero section with a cinematic food image, minimalist navigation, and elegant typography that emphasizes the theme of decoding sensory signatures through machine learning. The interface also provides an **"Initialize Mapping"** entry point, guiding users into the recommendation system.

• Query Interface (web.html)
The query module acts as the main interaction layer where users provide restaurant input. It features a clean and high-focus search bar within the **Inference Studio** interface. The system incorporates an AJAX-powered asynchronous autocomplete mechanism that retrieves real-time typeahead suggestions from the Pandas dataset index, helping reduce typographical errors and improving query accuracy. Floating restaurant visuals and a minimal dark interface further enhance the user experience.

• Inference Dashboard (result.html)
After a user submits a restaurant query, the backend processes the request using the precomputed **Cosine Similarity** model and displays results through a structured recommendation dashboard. Key features include:
• **Visual Analytics:** Integration of Chart.js is used to generate interactive visualizations, including a **Radar Chart** for rating comparison ("Palate Radar") and a **Polar Area Chart** for cost comparison ("Cost Polar Map").
• **Dynamic Recommendation Results:** A structured table dynamically displays the top recommended restaurants along with detailed metadata, including cuisine profile, rating score, and estimated cost for two people.
• **Similarity-Driven Output:** The dashboard presents a ranked list of highly similar restaurants generated through the recommendation engine, ensuring both relevance and quality.
• **Interactive Data Presentation:** Recommendations are enhanced with supporting graphical represe ntations, allowing users to interpret rating and cost relationships visually alongside tabular results.

Overall, the interface combines machine learning functionality with a premium user experience, making the recommendation process both practical and visually engaging.

a) Output Screenshots

The user journey is captured across three distinct visual states:

Figure 8.1: Vellore and Vine Landing Page (index.html)

The application opens to a premium landing interface detailing the core "Flavor Intelligence" logic. It features a hero-gradient layout with high-resolution imagery and sophisticated CSS animations (pinging status indicators and floating effects).

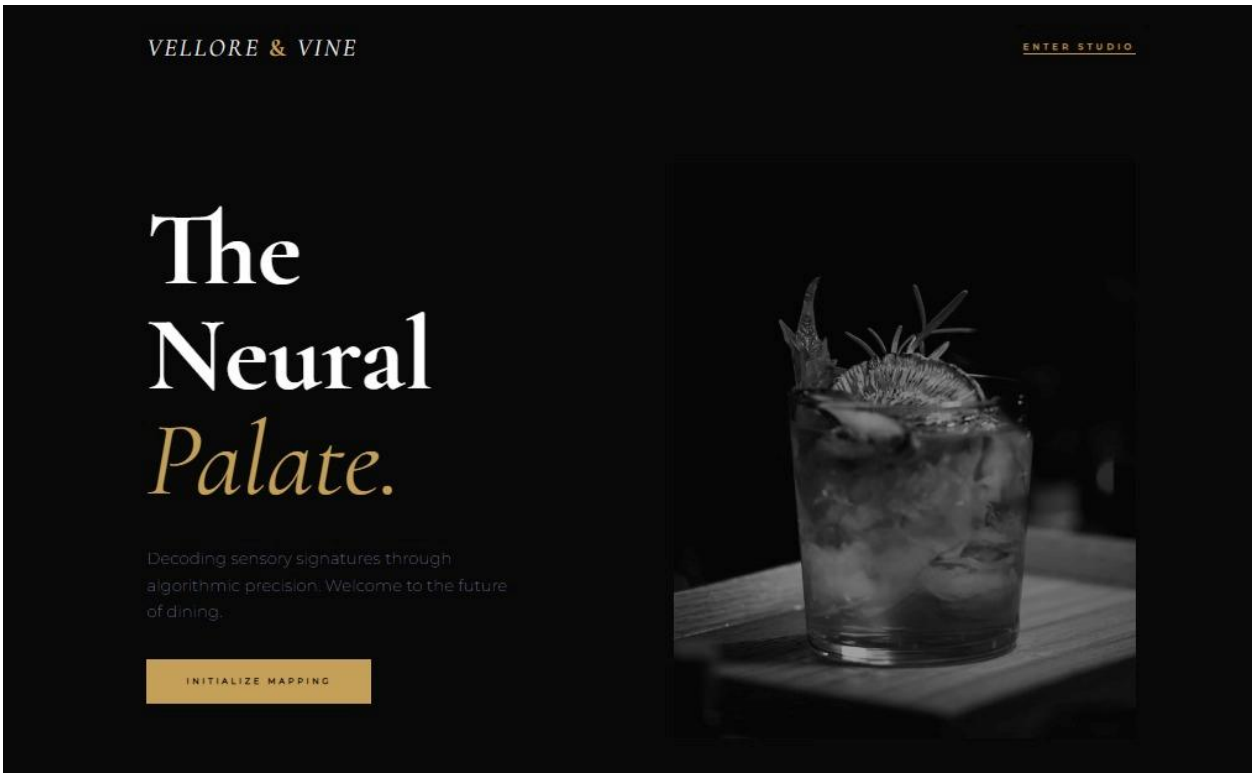
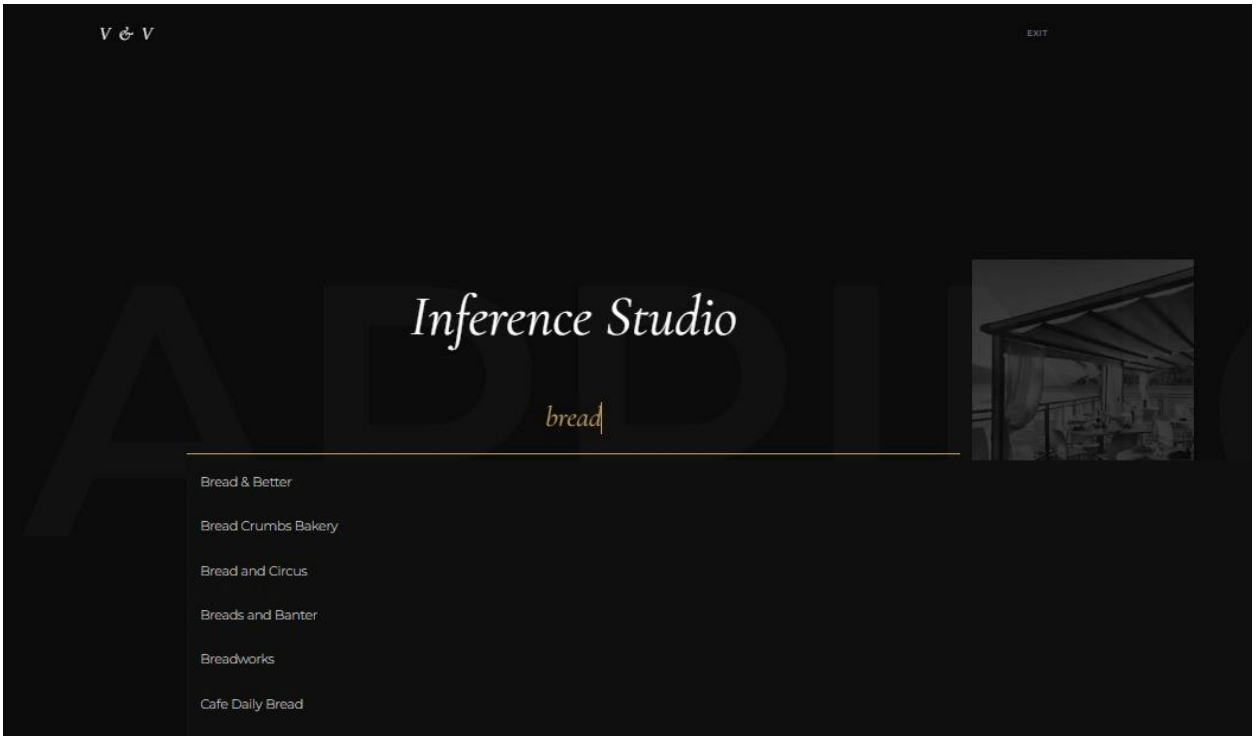


Figure 8.2: Vellore and Vine Query Terminal Interface (web.html)

The main interaction hub provides a clean, focused form for entering restaurant names. The input field features a dynamic **glow effect** and a **live suggestion dropdown** that allows users to select exact matches from the dataset before launching the inference engine.


V & V
EXIT
Inference Studio
bread
Bread & Better
Bread Crumbs Bakery
Bread and Circus
Breads and Banter
Breadworks
Cafe Daily Bread

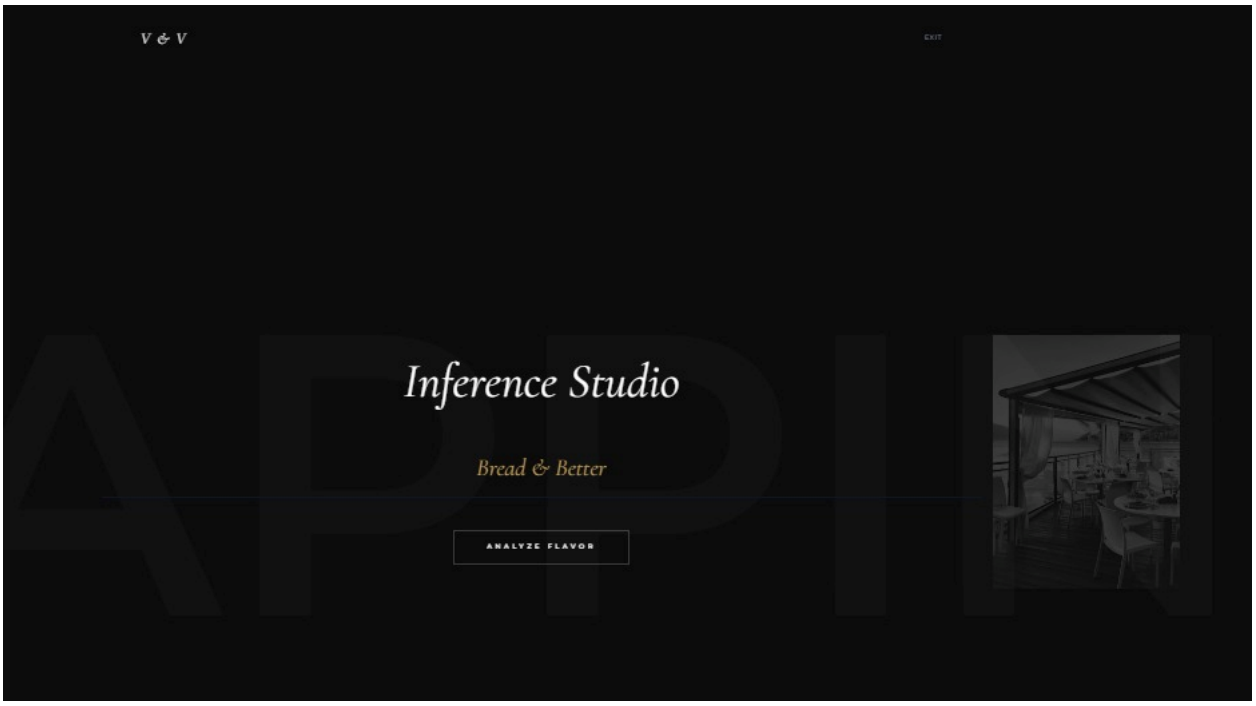

V & V
EXIT
Inference Studio
Bread & Better
ANALYZE FLAVOR

Figure 8.3: Recommendation Inference Dashboard (result.html)

The final dashboard confirms the NLP model's performance. It displays the Recommendation Dashboard containing a summary of the search term, match confidence bars, and an interactive results table. The bottom section features data visualizations that allow users to compare the "Cost for Two" and "Average Ratings" of the recommended venues at a glance.

9. Advantages and Disadvantages

a. Advantages

- **High Personalization:** Captures the actual "vibe" and specific culinary focus of a restaurant through text reviews and the "Flavor Intelligence" engine, rather than just suggesting a highly-rated but potentially irrelevant place.

- **User-Independent (No Cold Start):** Because the system is Content-Based, it does not require a user to have a long history or profile to receive excellent recommendations immediately.

- **Fast & Accessible:** The deployment via Flask provides a premium, responsive UI that abstracts the complex Cosine Similarity mathematics into a simple, high-end search bar.

- **Visual Insight**: Unlike standard lists, the system utilizes Chart.js to provide users with immediate visual comparisons of restaurant ratings and cost alignment.

b. Disadvantages

- **Item Cold-Start:** New restaurants with no reviews or metadata cannot be accurately vectorized and will not be recommended until sufficient text data is accumulated.

- **Data Dependency:** The accuracy of the similarity vectors is strictly bound to the quality and volume of the underlying text data provided in the dataset.

10. Conclusion

The **Vellore and Vine** project successfully merges advanced Natural Language Processing with luxury-tier full-stack web development. By transforming unstructured customer reviews into a mathematically comparable format, the engine automates the complex task of culinary curation.

Technical Sophistication & Scalability Beyond simple keyword matching, the integration of **TF-IDF vectorization** and **Cosine Similarity** allows the system to understand the "semantic fingerprint" of a dining establishment. This ensures that recommendations are based on the nuanced language of patrons—capturing elements like "ambient lighting," "umami depth," or "molecular technique"—which traditional category filters often overlook.

Bridge Between Data and Design The transition from raw data collection through to a fully interactive Flask application—complete with AJAX-powered search and real-time data visualization via **Chart.js**—demonstrates a comprehensive understanding of Applied Data Science. The project proves that complex machine learning models do not have to remain confined to scripts; they can be democratized through high-fidelity, user-centric interfaces that prioritize both performance and visual storytelling.

Operational Efficiency The resulting application is a fast, accurate, and professional tool that redefines the dining discovery experience. By utilizing pre-computed mathematical matrices (restaurant.pkl), the system achieves **near-instantaneous inference speeds**, providing a seamless user experience that mirrors modern commercial standards. This architecture sets a strong foundation for future AI-driven hospitality solutions, showcasing how data-driven intelligence can enhance the lifestyle and luxury service sectors.

11. Future Scope

While the current system is robust, several enhancements could be made in future iterations:

- **Live API Integration:** Connecting the backend to live APIs to dynamically update the TF-IDF matrix with fresh reviews daily.

- **Hybrid Implementation:** Adding an interactive map interface to allow users to narrow down mathematical recommendations by their immediate GPS radius.

- **Geospatial & Filter Upgrades:** Adding an interactive map interface and slider filters to allow users to narrow down the mathematical recommendations by their immediate GPS radius and specific budget constraints.

- **Advanced Personalization**: Utilizing the "Inference Engine" to allow users to input specific flavor profiles (e.g., "spicy and aromatic") rather than just existing restaurant names.

12. Appendix

a. Source Code

The project architecture consists of the following key files:

- **Restaurant_Recommendation_System.ipynb**: The primary Jupyter Notebook handling all EDA, text cleaning, vectorization, and serialization of the model.
- **restaurant1.csv**: The cleaned and formatted dataset used by the Flask application.
- **restaurant.pkl**: The serialized TF-IDF/Cosine Similarity model used for real-time inference.
- **app1.py**: The main Flask controller. It handles routing (/, /web, /result), processes AJAX autocomplete requests, and serves the recommendation dashboard.
- **templates/**: This directory contains the premium HTML5 files:
- **index.html**: The luxury-themed landing page.
- **web.html**: The search terminal featuring AJAX-powered restaurant autocomplete.
- **result.html**: The data-driven dashboard featuring Chart.js visualizations and recommendation tables.
- **static/**: Contains the branding assets, including the "Vellore and Vine" logo and custom CSS for glass-morphism effects.

b. GitHub & Project Video Demo Link

GitHub Repository Link: **https://github.com/IsrarAG/Restaurant_Recommendation-.git**

Project Video Demo Link:
https://drive.google.com/file/d/1fNJ1-cvobut0Hk1woRkpg4W_o0KVpMU1/view?usp=drivesdk